

Worksheet Problems on Data Structures and Algorithms

Worksheet 9

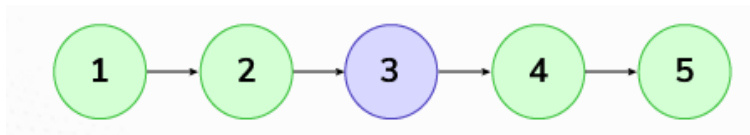
Q1: Middle of a Linked List

Given the head of a linked list, the task is to find the middle. For example, the middle of 1->2->3->4->5 is 3. If there are two middle nodes (even count), return the second middle. For example, middle of 1->2->3->4->5->6 is 4.

Examples:

Input: Linked list: 1->2->3->4->5

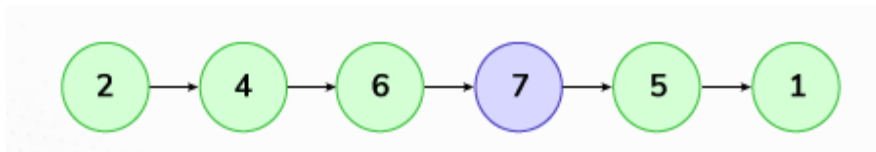
Output: 3



Explanation: The given linked list is 1->2->3->4->5 and its middle is 3.

Input: Linked list: 2->4->6->7->5->1

Output: 7



Explanation: The given linked list is 2->4->6->7->5->1 and its middle is 7.

Q2: Reverse a linked list

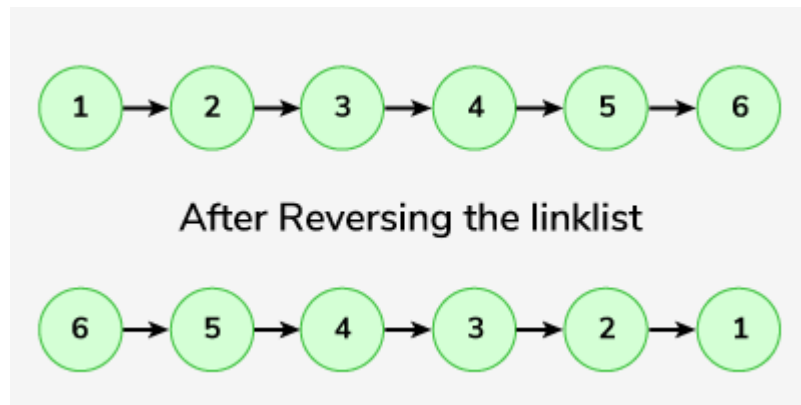
Given the head of a linked list, the task is to reverse this list and return the reversed head.

Examples:

Input: Linked list: 1->2->3->4->5->6

Output: 6->5->4->3->2->1

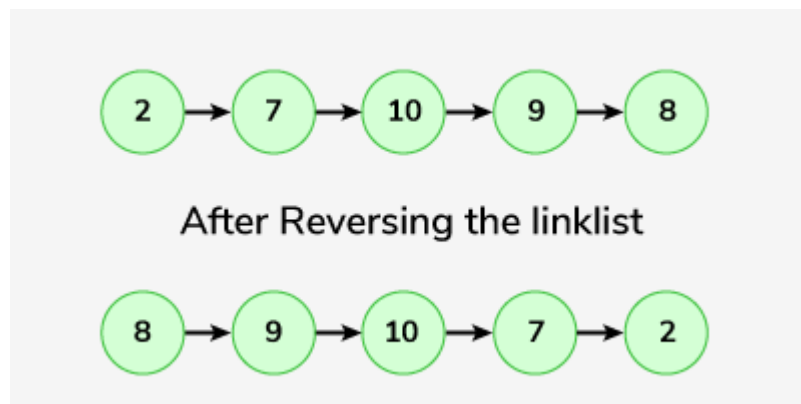
Explanation:



Input: Linked list: 2->7->10->9->8

Output: 8->9->10->7->2

Explanation:



Input: Linked List: 10

Output: 10

Explanation: For a single node list, the reverse would be same as original

Q3: Rotate a Linked List

Given the head of a singly linked list, the task is to rotate the left shift of the linked list by k nodes, where k is a given positive integer smaller than or equal to the length of the linked list.

Examples:

Input: linkedlist: 2->4->7->8->9 , $k = 3$

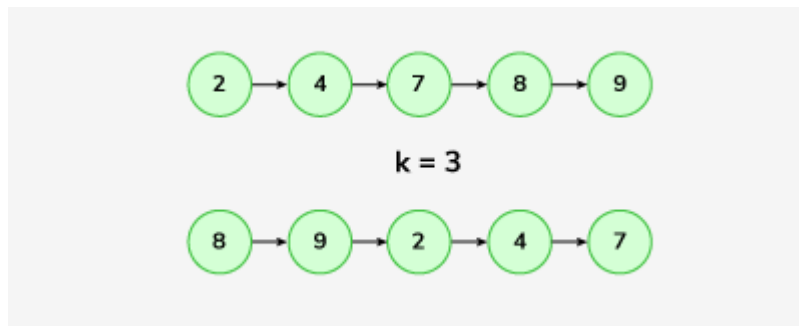
Output: 8->9->2->4->7

Explanation:

Rotate 1: 4 -> 7 -> 8 -> 9 -> 2

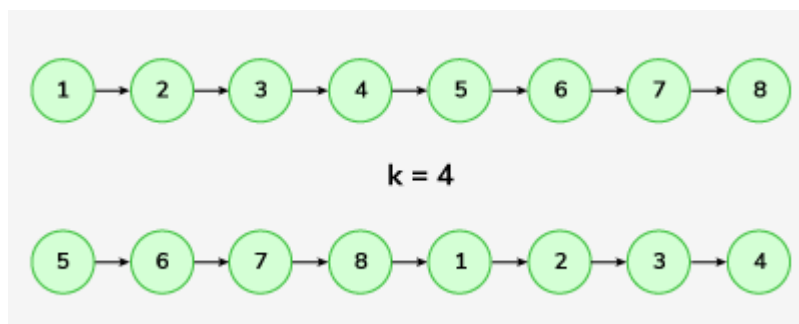
Rotate 2: 7 -> 8 -> 9 -> 2 -> 4

Rotate 3: 8 -> 9 -> 2 -> 4 -> 7



Input: linkedlist: 1->2->3->4->5->6->7->8 , $k = 4$

Output: 5->6->7->8->1->2->3->4



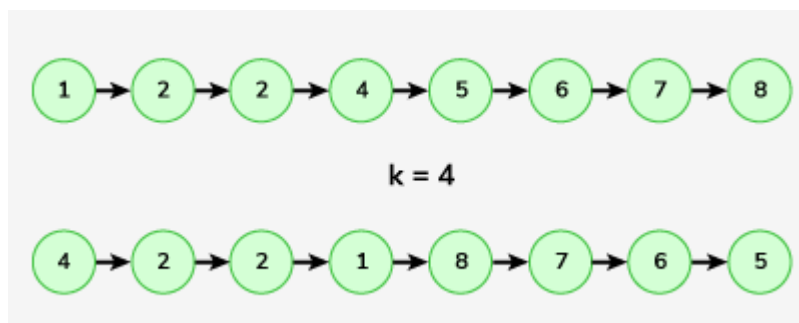
Q4: Reverse a Linked List in Groups

Given a linked list, the task is to reverse every k node (where k is an input to the function) in the linked list. If the number of nodes is not a multiple of k then left-out nodes, in the end, should be considered as a group and must be reversed (See Example 2 for clarification).

Examples:

Input: Linked List: 1->2->2->4->5->6->7->8, $k = 4$

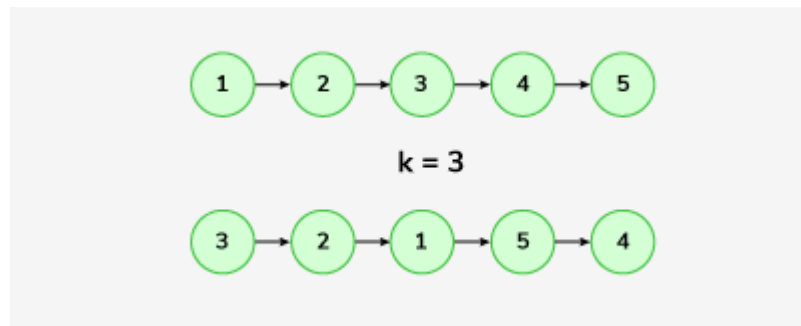
Output: 4 -> 2 -> 2 -> 1 -> 8 -> 7 -> 6 -> 5



Explanation: The first 4 elements 1,2,2,4 are reversed first and then the next 4 elements 5,6,7,8. Hence, the resultant linked list is 4->2->2->1->8->7->6->5.

Input: LinkedList: 1->2->3->4->5, $k = 3$

Output: 3 -> 2 -> 1 -> 5 -> 4



Explanation: The first 3 elements 1,2,3 are reversed first and then element 4,5 are reversed. Hence, the resultant linked list is 3->2->1->5->4.

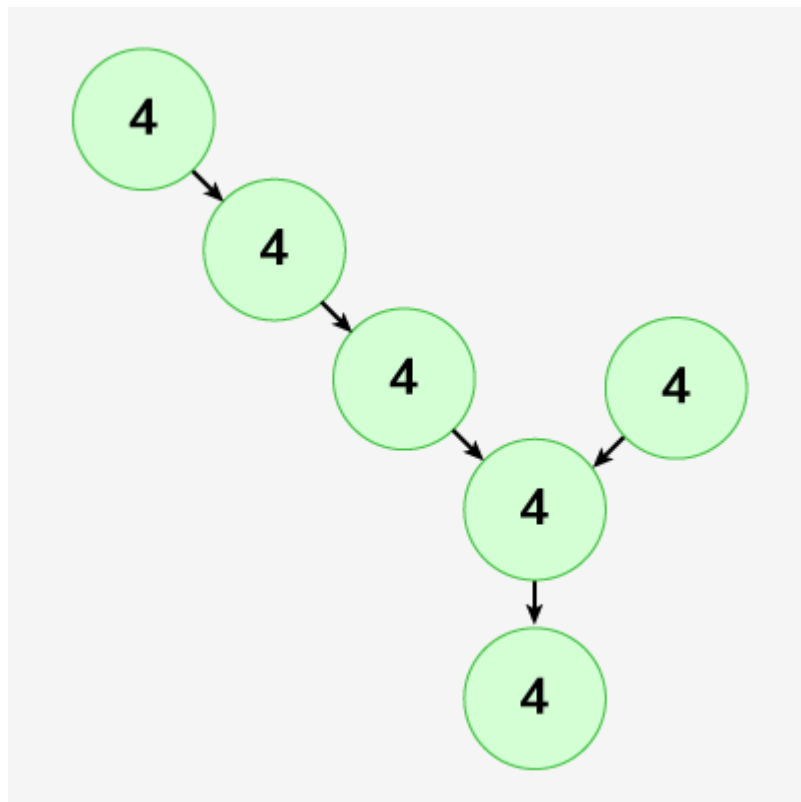
Q5: Intersection Point in Y Shaped Linked Lists

Given two singly linked lists, return the point where two linked lists intersect.

Note: If the linked lists do not merge at any point, return -1.

Examples:

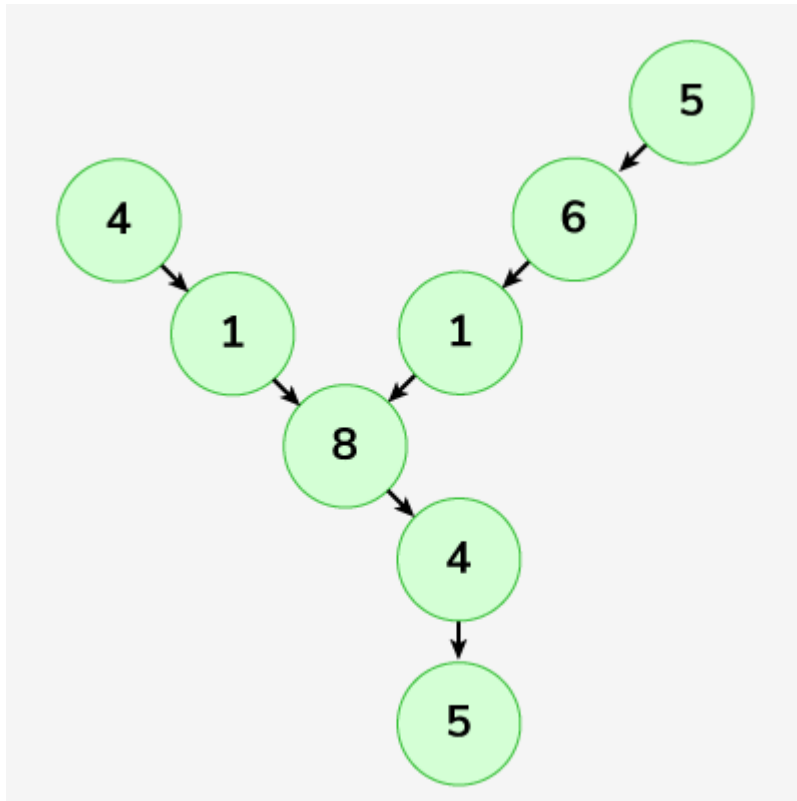
Input: Linked list 1: 4->4->4->4->4, Linked list 2: 4->4->4



Output: 4

Explanation: From the above image, it is clearly seen that the common part is 4->4 whose starting point is 4.

Input: Linked list 1: 4->1->8->4->5, Linked List 2: 5->6->1->8->4->5



Output: 8

Explanation: From the above image, it is clearly seen that the common part is 8->4->5 whose starting point is 8.

Input: Linked list 1: 1->2->3, Linked list 2: 4->5->6

Output: -1

Explanation: There is no common part, so there is no intersection point.